

Arithmetic and Order

- **Definition 1**

The set of **integers** is

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

In this course, we will consider that 0 is not a natural number, that is, the set of **natural numbers** is

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

We endowed the set $\mathbb{Z}$ with two binary operations

$$\begin{aligned} + : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} & \cdot : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (m, n) &\mapsto m + n & (m, n) &\mapsto m \cdot n \end{aligned}$$

The arithmetic of $\mathbb{Z}$ is generated by the properties that these operations satisfy.

Notation: $m \cdot n = mn$.

- **Properties 1.1**

1. If $m, n, r \in \mathbb{Z}$ then

$$(m + n) + r = m + (n + r). \quad \text{Associativity of sum}$$

2. There exists a unique element $0 \in \mathbb{Z}$ such that for all $m \in \mathbb{Z}$

$$m + 0 = m = 0 + m. \quad \text{Existence of identity for the sum}$$

3. For each $m \in \mathbb{Z}$ there exists a unique element $-m \in \mathbb{Z}$ such that

$$m + (-m) = 0 = (-m) + m. \quad \text{Existence of inverse for the sum}$$

4. If $m, n \in \mathbb{Z}$ then

$$m + n = n + m. \quad \text{Commutativity for the sum}$$

5. If $m, n, r \in \mathbb{Z}$ then

$$(mn)r = m(nr). \quad \text{Associativity of product}$$

6. There exists a unique $1 \in \mathbb{Z}$ such that for all $m \in \mathbb{Z}$

$$m \cdot 1 = m = 1 \cdot m. \quad \text{Existence of identity for the product}$$

7. If $m, n, a \in \mathbb{Z}$ with $a \neq 0$ and $am = an$ then

$$m = n. \quad \text{Cancellation law}$$

8. If $m, n \in \mathbb{Z}$ then

$$mn = nm. \quad \text{Commutativity for the product}$$

9. If $m, n, r \in \mathbb{Z}$ then

$$m(n + r) = mn + mr. \quad \text{Distributivity of product}$$

- **Proposition 1.2**

$m \cdot 0 = 0$ for all $m \in \mathbb{Z}$.

- **Proof 1.2.1**

$$\begin{aligned} m \cdot 0 &= m \cdot (0 + 0) && \text{Property 2} \\ &= m \cdot 0 + m \cdot 0 && \text{Property 9} \end{aligned}$$

hence

$$\begin{aligned} m \cdot 0 + (-m \cdot 0) &= (m \cdot 0 + m \cdot 0) + (-m \cdot 0) \\ 0 &= m \cdot 0 + (m \cdot 0 + (-m \cdot 0)) && \text{Property 1, 3} \\ 0 &= m \cdot 0 + 0 && \text{Property 3} \\ 0 &= m \cdot 0. && \text{Property 2} \end{aligned}$$

◦ **Proposition 1.3**

For all $m, n \in \mathbb{Z}$, if $mn = 0$ then $m = 0$ or $n = 0$.

■ **Proof 1.3.1**

Let $m, n \in \mathbb{Z}$ such that $mn = 0$. If $m = 0$, there is nothing to prove.

If $m \neq 0$, then

$$mn = 0 = m \cdot 0. \quad \text{Proposition 1.2}$$

and by Property 7 we have that $n = 0$.

◦ **Proposition 1.4**

$-m = (-1) \cdot m$ for all $m \in \mathbb{Z}$.

■ **Proof 1.4.1**

Since

$$\begin{aligned} m + (-1)m &= m \cdot 1 + m \cdot (-1) && \text{Property 6, 8} \\ &= m(1 + (-1)) && \text{Property 9} \\ &= m \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

therefore $(-1)m$ is an additive inverse of m and by the uniqueness of the additive inverse we have $-m = (-1)m$.

◦ **Proposition 1.5**

$-(-m) = m$ for all $m \in \mathbb{Z}$.

■ **Proof 1.5.1**

Since $m + (-m) = 0$, the integer m is the additive inverse of $-m$ and by the uniqueness of the additive inverse we have that $m = -(-m)$.

◦ **Proposition 1.6 (Rules for signs)**

For all $m, n \in \mathbb{Z}$,

$$(-m)n = -(mn), \quad m(-n) = -(mn), \quad (-m)(-n) = mn.$$

■ **Proof 1.6.1**

$$\begin{aligned} 1. \quad mn + (-m)n &= nm + n(-m) && \text{Property 8} \\ &= n(m + (-m)) && \text{Property 9} \\ &= n \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

Therefore $(-m)n$ is an additive inverse of mn and by the uniqueness of the additive inverse we have $(-m)n = -(mn)$.

$$\begin{aligned} 2. \quad mn + m(-n) &= m(n + (-n)) && \text{Property 9} \\ &= m \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

Therefore $m(-n)$ is an additive inverse of mn and by the uniqueness of the additive inverse we have $m(-n) = -(mn)$.

$$\begin{aligned} 3. \quad & (-m)(-n) = -(m(-n)) && \text{Proposition 1.6(1)} \\ & = -(-mn) && \text{Proposition 1.6(2)} \\ & = mn. && \text{Proposition 1.5} \end{aligned}$$

• **Remark 1**

If $m^2 = n^2$ it does not imply that $m = n$, for example, $2^2 = 4 = (-2)^2$ but $2 \neq -2$. In general

$$0 = m^2 - n^2 = (m - n)(m + n).$$

And by the previous proposition we have that $m - n = 0 \vee m + n = 0$, that is, $m = n \vee m = -n$.

• **Definition 2**

The **order** in $\mathbb{Z}$ is defined in the following form

$$m \leq n \iff n = m + k \text{ for some } k \in \mathbb{N}_0.$$

Equivalently,

$$m \leq n \iff n - m \in \mathbb{N}_0.$$

The **strict order** in $\mathbb{Z}$ is defined in the following form

$$m < n \iff m \leq n \wedge m \neq n.$$

The relation $\leq$ is a **total order** in $\mathbb{Z}$, that is, reflexive, antisymmetric, transitive and any two integers can be compared.

◦ **Properties 2.1**

1. For all $m, n \in \mathbb{Z}$ either $m \leq n$ or $n \leq m$.
2. If $m, n, r \in \mathbb{Z}$ with $m \leq n$ then $m + r \leq n + r$.
3. If $m, n, r \in \mathbb{Z}$ with $m \leq n \wedge r \geq 0$ then $mr \leq nr$.

• **Definition 3**

The **absolute value** of an integer m is

$$|m| = \begin{cases} m & (m \geq 0) \\ -m & (m < 0) \end{cases}.$$

◦ **Proposition 3.1**

Let $m, n \in \mathbb{Z}$,

1. $m \leq n$ if and only if $-n \leq -m$.
2. $m^2 \geq 0$.
3. $m^2 = 0$ if and only if $m = 0$.

■ **Proof 3.1.1**

1. ($\implies$) Suppose that $m \leq n$ then by Property 2

$$\begin{aligned} m - (m + n) &\leq n - (m + n) \\ -n &\leq -m. \end{aligned}$$

- ($\impliedby$) Suppose that $-n \leq -m$, then by Property 2

$$\begin{aligned} -n + (n + m) &\leq -m + (n + m) \\ m &\leq n. \end{aligned}$$

2. If $m \geq 0$ then by Property 3

$$m^2 = m \cdot m \geq m \cdot 0 = 0.$$

If $m \leq 0$ then by Proposition 3.1(1) $-m \geq 0$ and by Property 3 we have

$$m^2 = (-m)(-m) \geq (-m) \cdot 0 = 0.$$

3. ($\Rightarrow$) If $m^2 = 0$ then $m \cdot m = m^2 = 0$ and thus $m = 0$.
 ($\Leftarrow$) If $m = 0$ then $m^2 = m \cdot m = 0 \cdot 0 = 0$.

• **Definition 4**

A integer n is **even** if $n = 2k$ for some $k \in \mathbb{Z}$.

A integer n is **odd** if $n = 2k + 1$ for some $k \in \mathbb{Z}$.

◦ **Example 4.1**

-10 is even since $-10 = 2 \cdot (-5)$.

7 is even since $7 = 2 \cdot 3 + 1$.

◦ **Example 4.2**

Prove that the sum and product of two even integers is even.

■ **Proof 4.2.1**

Suppose that n, n' are even. Then by the definition

$$n = 2k, n' = 2k' \text{ for some } k, k' \in \mathbb{Z}$$

hence

$$\begin{aligned} n + n' &= 2k + 2k' = 2(k + k') \text{ with } k + k' \in \mathbb{Z} \\ n \cdot n' &= (2k) \cdot (2k') = 2 \cdot (2kk') \text{ with } 2kk' \in \mathbb{Z} \end{aligned}$$

Thus that the sum and product of two even integers is even.

◦ **Exercise 4.3**

Are the sum and product of two odd integers odd?

■ **Solution 4.3.1**

Suppose that n, n' are odd. Then by the definition

$$n = 2k + 1, n' = 2k' + 1 \text{ for some } k, k' \in \mathbb{Z}$$

hence

$$\begin{aligned} n + n' &= (2k + 1) + (2k' + 1) = 2(k + k' + 1) \text{ with } k + k' + 1 \in \mathbb{Z} \\ n \cdot n' &= (2k + 1) \cdot (2k' + 1) = 2 \cdot (2kk' + k + k') + 1 \text{ with } 2kk' \in \mathbb{Z} \end{aligned}$$

Thus the product of two odd integers is odd but the sum of them is even.